

Data Science Program, George Washington University

Sample Exam 1

DATS 6103 (Introduction to Data Mining)

Date: **October 27, 2026**

Instructor: Prof. Sushovan Majhi

Name: _____

Please read the following instructions carefully.

- This is a **sample paper** for practice only, at the real length and the real difficulty. It contains 18 questions for a total of 100 points, covering everything through Oct 20, but **not** regression as prediction.
- You have **2 hours 30 minutes**, the full session. Answer all questions.
- The exam is **open-book** and **open-note**, but **printed material only**: the chapters covered, the labs whose solutions have been released, the homework covered with its solutions, and your own notes.
- **Laptops, tablets, phones** and **watches** must be **switched off and put away**. No AI assistance and no communication of any kind.
- **Nothing has to be executed**. Code you write is graded on whether it would do the right thing, not on a missing colon; a shape, a dtype or the name of an error earns marks.
- All notation has its usual meaning. You may use standard results directly by stating them appropriately. **Read the instructions for individual questions carefully** before answering.

Question	Points	Score
1	6	
2	5	
3	4	
4	8	
5	7	
6	6	
7	4	
8	5	
9	5	
10	5	
11	6	
12	5	
13	5	
14	6	
15	4	
16	4	
17	10	
18	5	
Total:	100	

Part A Python

1. Say what each prints, or name the error.

(a) (2 points)

```
1 a = [1, 2, 3]
2 b = a
3 b.append(4)
4 print(len(a))
```

(b) (2 points)

```
1 def tag(item, seen=[]):
2     seen.append(item)
3     return seen
4 tag("x")
5 print(tag("y"))
```

(c) (2 points)

```
1 print([n * 2 for n in range(4)][-1])
```

2. (5 points) This function is meant to return a cleaned copy and leave its argument alone. It does not. Say what goes wrong, and write the corrected body.

```
1 def drop_negatives(values):
2     for i, v in enumerate(values):
3         if v < 0:
4             values[i] = 0
5     return values
```

3. (4 points) $d = \{"b": 2, "a": 1\}$. Say what `list(d)` gives and why, then what `sorted(d)` gives.

Part B NumPy

4. For each pair of shapes, give the shape of $A + B$, or write **raises**.

(a) (2 points) (30, 12, 4) and (4,)

(b) (2 points) (30, 12, 4) and (12,)

(c) (2 points) (30, 1, 4) and (12, 1)

(d) (2 points) (3, 4) and (4, 3)

5. T has shape (30, 12, 4): 30 days, 12 sensors, 4 channels.

(a) (3 points) What shape does `T.mean(axis=1)` return, and what does each number mean?

(b) (2 points) Write the expression for the mean of each **channel** across everything else.

(c) (2 points) What does `T.mean(axis=(0, 1), keepdims=True)` return, and why would you want `keepdims`?

6. For each, say whether the result **shares memory** with T.

- (a) (1 point) `T[0]`
- (b) (1 point) `T.reshape(-1, 4)`
- (c) (1 point) `T[T[..., 0] > 25]`
- (d) (1 point) `T.T`

(e) (2 points) In one sentence: what do the ones that share have in common?

7. (4 points) `A = np.arange(6).reshape(2, 3)`. Give A, then `A.T`, then say what `A.ravel()` and `A.T.ravel()` each produce.

Part C Linear algebra in NumPy

8. (5 points) A is $(15, 4)$, x is $(4,)$, y is $(15,)$. For each, give the resulting shape or write **raises**: $A @ x$, $A.T @ y$, $A @ A$, $A.T @ A$, $A @ A.T$.

9.

(a) (2 points) Why is `np.linalg.solve(A, b)` preferred to `np.linalg.inv(A) @ b`?

(b) (3 points) If $\text{cond}(A) = 10^4$, what is $\text{cond}(A^T A)$, and what does that cost you in accuracy?

10. X is $(40, 6)$ and `U, S, Vt = np.linalg.svd(X, full_matrices=False)`.

(a) (2 points) Give the shapes of U , S and Vt .

(b) (1 point) Write the expression that reconstructs X .

(c) (2 points) Is $U @ U.T$ the identity? Is $U.T @ U$?

SAMPLE

Part D Pandas

11.

```
1 q1 = pd.Series([100, 120, 90], index=["shaw", "petworth", "navy yard",  
   ""])  
2 q2 = pd.Series([110, 95, 130], index=["petworth", "navy yard", "  
   brookland"])  
3 total = q1 + q2
```

(a) (3 points) How many entries does `total` have, and how many are `NaN`?

(b) (1 point) Give `total["petworth"]`.

(c) (2 points) Write the expression that treats a store absent from a quarter as zero.

12. (5 points) `s = pd.Series([10, 20, 30, 40], index=[2, 3, 4, 1])`. Give `s.loc[2]` and `s.iloc[2]`, and say in one sentence why `s[2]` is a bad habit even though it returns something here.

13.

```
1 ride = pd.DataFrame({"week1": [12.0, 8.0, np.nan],  
2                        "week2": [15.0, 9.0, 21.0]},  
3                        index=["a", "b", "c"])
```

(a) (2 points) Give `ride.mean(axis="index")`.

(b) (1 point) Give `ride.mean(axis="columns")["c"]`.

(c) (2 points) Give `ride.mean(axis="index", skipna=False)["week1"]`, and say why the default is dangerous.

14.

```
1 orders = pd.DataFrame({"order_id": [1, 2, 3], "cust": ["a", "b", "c"],  
2                        "total": [30, 55, 12]})  
3 lookup = pd.DataFrame({"cust": ["a", "b", "b", "c"],  
4                        "segment": ["retail", "retail", "wholesale", "  
5                        retail"]})  
5 merged = orders.merge(lookup, on="cust")
```

(a) (2 points) How many rows does merged have?

(b) (2 points) What is `merged["total"].sum()`, and what was it before the merge?

(c) (2 points) Name the argument that would have made the merge complain instead.

15. (4 points)

```
1 orders = pd.DataFrame({"size": ["S", "M", "L", "XL", "M"]})
2 sizes = {"S": 1, "M": 2, "L": 3}
3 coded = orders["size"].map(sizes)
```

Give coded, its dtype, and what a later `coded.mean()` would be computed over.

16.

```
1 tickets = pd.DataFrame({"team": ["core", "core", "web"],
2                             "week": ["w1", "w1", "w1"],
3                             "hours": [3, 5, 2]})
```

(a) (2 points) What does `tickets.pivot_table(index="team", columns="week", values="hours")` give for core in w1?

(b) (2 points) What would `pivot()` have done?

Part E Exploratory analysis

17. A colleague sends a histogram of 3,000 rents with five bins and concludes the distribution is a single mound centred on \$2,000.
- (a) (4 points) Give two distinct reasons the conclusion may not survive a different chart.
 - (b) (3 points) Name the summary statistic you would ask for alongside it, and what you would learn.
 - (c) (3 points) They then report the mean rent as \$2,180 and the median as \$1,740. What do you now know about the shape?

Part F Statistical learning

18.

- (a) (3 points) A model scores 0.02 mean squared error on the data it was fitted to and 0.35 on data it has not seen. Name what has happened, and what the 0.02 is worth.
- (b) (2 points) In one sentence, what does making a model more flexible do to bias, and to variance?

*Code is graded on whether it would do the right thing.
A shape, a dtype or the name of an error earns marks.*